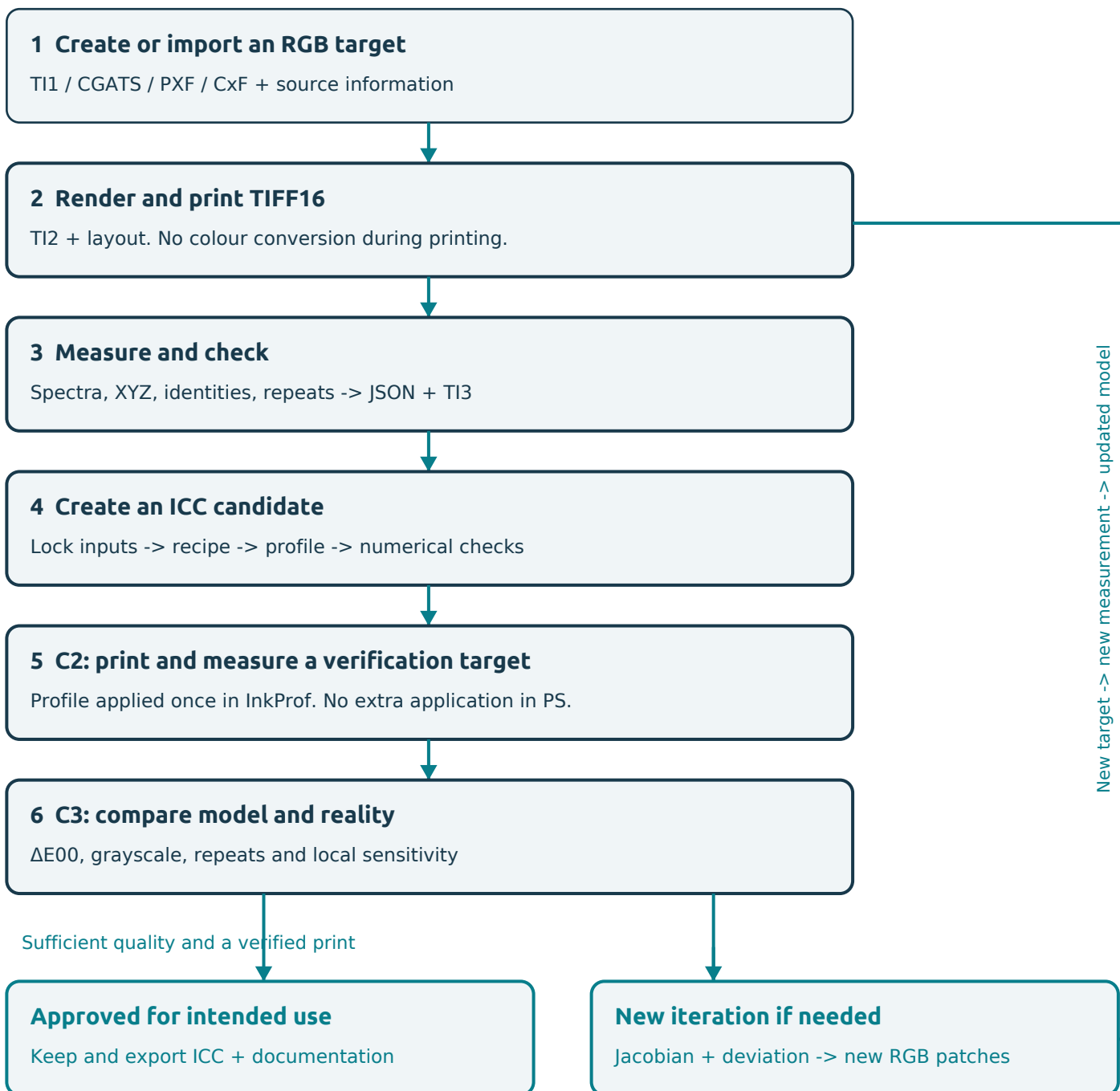


From RGB patches to ICC

A practical guide to targets, measurement, profiling, verification and feedback.

Publisher: Christer Törnkvist, christer@borgasundsfotografiska.se



Address measurement errors first: remeasure suspect rows or patches before allowing them to guide the model. Not every small change in $\Delta E00$ automatically calls for more patches.

Throughout the workflow: a shared project folder, JSON, manifests, source files and logs. Keep training points, development points and controls separate.

1. Project and print targets

The examples are separate steps. Finish the previous dialog and measurement before proceeding.

Start InkProf and create a new project

```
cd('/Users/christer/Desktop/InkProf')
paths = setupInkProf();
rehash
projectFolder = fullfile(paths.Projects,'New-paper');
inkprof.createProject(projectFolder,Name="New paper");
```

Input: a new project folder. Do not run createProject again for an existing project; simply set projectFolder to its existing path.

Choose a method and render in two separate windows

```
inkprof.designTarget();
inkprof.renderTarget();
```

designTarget: enter a name, method (InkProf mesh or Argyll OFPS), point budget, gray samples and controls. Save the definition as TI1 + JSON. renderTarget: select this TI1 or an imported RGB definition; set DPI, paper and output file. Check the page count and preview.

A4 landscape is the default in the TIFF window. The maximum width is 320 mm; the user chooses the length. The printed patch order and coordinates are saved in the accompanying TI2 and JSON.

Alternative: create an Argyll target directly

```
targetFolder = fullfile(projectFolder,'targets','base-target');
inkprof.createTarget(targetFolder,PatchCount=575, ...
    Paper="A4-landscape",DPI=300,SpacerMode="bw", ...
    Randomize=true,Seed=42);
```

Input: a new output folder and patch count. For an existing target, use Source=sourceFile instead of PatchCount, where sourceFile is the full path to the definition. RGB is supported; CMYK must not be interpreted as RGB.

Output: TIFF16, target.ti2 and layout metadata. Print at 100%, with contrast markers and no colour conversion. Keep the same paper, ink and driver settings throughout profiling.

2. Measurement and external files

TI2 describes the printed sheet. Measurements are normalised to InkProf's internal JSON.

Measure the target that was actually printed

```
ti2File = fullfile(targetFolder,'target.ti2');
stamp = string(datetime('now','Format','yyyyMMdd-HH:mm:ss'));
sessionFolder = fullfile(projectFolder,'measurements',stamp);
dialog = inkprof.measureChart(ti2File, ...
    SessionFolder=sessionFolder,ScanMode="paired", ...
    Condition="M0",PairedWarningDeltaE=1);
```

Input: TI2 matching the printed patches, order and layout exactly. Calibrate as instructed by the dialog. Hold the instrument button during the scan and follow the row and page indicators. Save when measurement is complete.

ScanMode="single": one scan per row. "alternating": the next row is scanned in the opposite direction. "paired": forward and backward scans of the same row, followed by averaging and a check of their difference.

Output: measurement-*.json, measurement data in TI3 and session records. After any remeasurements, select the latest accepted revision rather than an older original.

Select a saved measurement revision for the next step

```
[name,where] = uigetfile('*.json','Select measurement revision');
if isequal(name,0), return; end
measurementFile = fullfile(where,name);
```

Select measurement-*.json. proposal.json, manifests, chart reports and verification.json are different documents and must not be selected as measurements.

If the measurement comes from another application

```
[result,measurementFile,sessionFolder] = ...
    inkprof.importMeasurement();
```

Dialog input: MXF or TI3. TI3 requires matching target data (TI2), which can also be specified using TargetFile=ti2File. For MXF, available identities and layout information are used. Missing print or measurement conditions need to be supplied, not guessed.

Averaging does not replace quality control: review inconsistent scans before averaging and retain the raw data. Small differences, such as 0.6 versus 0.7 ΔE_{00} , can be diagnostically useful without implying a practically worse print.

3. Create and review an ICC profile

Choose the automatic workflow or the manual B steps. These are alternative routes.

Automatic workflow with traceable iteration

```
[iterationFolder,result] = inkprof.iterateProfile( ...
    measurementFile,ProjectFolder=projectFolder, ...
    Name="New paper - iteration 1", ...
    MaxNewPatches=100,NormTarget=1,GrayWeight=2);
jobFolder = fullfile(iterationFolder,result.selectedJob);
profileFile = fullfile(iterationFolder,result.profileFile);
verificationFolder = fullfile(iterationFolder, ...
    result.verification.folder);
```

Input: an accepted measurement and a project folder. Each patch has an assigned use: training, development or control (explained on page 8). Without this partition, the source patches are used for training, leaving no separate development set for candidate comparison. RoleFile specifies a file containing this partition.

Output: an ICC candidate, saved training inputs, a C2 target, iteration.json and progress.log/progress.jsonl. MaxNewPatches limits new patch sampling; it is not the number of iterations. The program does not automatically approve physical print quality.

Manual route: B1, B2 and B3

```
[inputFolder,input] = inkprof.prepareProfileInput( ...
    measurementFile,Name="New paper");
[recipeFile,recipe] = inkprof.createProfileRecipe(inputFolder);
[jobFolder,status] = inkprof.runProfileJob(recipeFile);
profileFile = fullfile(jobFolder,'result','profile.icc');
```

B1 locks a checked measurement revision. B2 takes this input folder and saves a recipe. B3 takes recipe.json and builds the ICC candidate. A successful job means the profile was created, not that the print has been verified.

Numerical checks before printing

```
[fit,fitFile] = inkprof.checkProfileFit(jobFolder);
[grid,gridFile] = inkprof.checkProfileGrid(jobFolder);
[c1,c1File] = inkprof.checkProfileC1(jobFolder);
```

Input: the job folder, not the ICC file itself. Fit measures the fit to training data. Grid/C1 examine the inverse, ramps and colour engines, among other things. These checks do not replace C2/C3 using independently printed samples.

4. C2 printing and C3 analysis

A new print and measurement test the profile's actual colour reproduction.

C2 already exists after iterateProfile

Use verificationFolder from step 3. After the manual B route, create C2 with the following call instead:

```
[verificationFolder,reference] = ...
    inkprof.createVerificationTarget(jobFolder, ...
    Name="C2",Paper="A4-landscape",DPI=300);
```

Input: a successful profile job folder. Output: verification.json, print/target.tif (and any additional pages), print/target.ti2 and metadata.

The profile is already applied to the C2 image's RGB values

C2 uses absolute colorimetric conversion without black point compensation in InkProf. The TIFF has no embedded profile, but its pixel values have already been converted. Do not apply the ICC profile again in Photoshop. The printing pipeline must leave these RGB values unchanged by any further colour conversion.

Record the application, driver, media type, quality, colour management and scale. Use the same physical paper and settings as for the profiling inputs. Allow the print to dry in the same way.

Measure and save in the same verification folder

```
stamp = string(datetime('now','Format','yyyyMMdd-HH:mm:ss'));
dialog = inkprof.measureChart( ...
    fullfile(verificationFolder,'print','target.ti2'), ...
    SessionFolder=fullfile(verificationFolder, ...
    'print',"measurement-"+stamp), ...
    ScanMode="paired",Condition="M0");
```

After saving, select the new measurement-*.json as on page 3 and name its path c2MeasurementFile. Do not select the original profiling measurement.

```
[report,reportFile] = inkprof.checkVerificationTarget( ...
    fullfile(verificationFolder,'verification.json'), ...
    c2MeasurementFile);
```

Input 1: the C2 reference. Input 2: the new C2 measurement. Output: verification-check.json and reports under checks. These describe target error, model versus measurement, grayscale and repeats, among other things. Internal thresholds do not certify ISO conformity.

5. Feedback and additional patches

Check measurements first. Only then use remaining deviations to guide the next sampling round.

Review the outcome and create a refinement target

```
[feedback,feedbackFile] = inkprof.analyseVerification(reportFile);
[proposal,proposalFolder] = inkprof.refineVerification( ...
    reportFile,Name="Iteration 2",MaxNewPatches=100, ...
    NormTarget=1,ErrorThreshold=1,GrayWeight=2, ...
    RadiusPercent=5,MinSpacingPercent=1,CreatePrint=true);
```

Input: the C3 report verification-check.json. Its sources must be traceable to the profile, reference and measurement. The Jacobian describes local model sensitivity; the deviation indicates where more data are needed. Near-singularity calls for cautious, bounded steps and control points.

NormTarget is the iteration routine's stopping target, not a promise of final $\Delta E00$. MaxNewPatches is an upper limit on new points; repeated control patches may be added beyond it. GrayWeight prioritises grayscale, but any colour may be selected.

Output: proposal.json, target.ti1, a log and new print inputs. This is a device-RGB target for model refinement: print without applying an ICC profile, unlike the already converted C2 target.

Measure the additional samples

```
stamp = string(datetime('now','Format','yyyyMMdd-HH:mm:ss'));
dialog = inkprof.measureChart( ...
    fullfile(proposalFolder,'refinement-print','print', ...
        'target.ti2'), ...
    SessionFolder=fullfile(proposalFolder,'measurements',stamp), ...
    ScanMode="paired",Condition="M0");
```

Link the new measurement to previous inputs

```
[nextIteration,result] = inkprof.continueRefinement( ...
    proposalFolder,newMeasurementFile, ...
    Name="Iteration 2",MaxNewPatches=100, ...
    NormTarget=1,GrayWeight=2);
verificationFolder = result.verificationFolder;
```

newMeasurementFile is the saved refinement measurement. The first argument is the folder containing proposal.json, not the JSON file. Without arguments, a folder picker opens, followed by a file picker. The link validates provenance, patches and their uses; a new ICC and C2 are created. MaxNewPatches now applies to the next proposal.

6. Files, decisions and reuse

Each file type has a specific purpose. A filename alone is not enough to link data reliably.

TI1 - RGB definition

Input to target rendering. Specifies the intended device-RGB values before printing.

TI2 - print and measurement layout

Input to measureChart/chartread. Must match the physical sheet, pagination and patch order.

TI3 / MXF - measured data

Imported with identities, spectra and available measurement conditions. TI3 needs matching target data.

measurement-*.json - accepted revision

Input to profiling or C3. Select the correct revision after remeasurement. The original is preserved.

verification.json / verification-check.json

The first is the C2 reference. The second is the C3 result and input to feedback/refinement.

proposal.json and its folder

Proposal provenance, new patches and their uses. continueRefinement takes the folder and new measurement file.

Read, import or export an ICC profile without changing it

```
[profile,jsonFile] = inkprof.readICC(profileFile);
[copyFile,receipt] = inkprof.saveICC(profileFile);
% Import an external profile into the project:
[importedFile,record] = inkprof.importICC(sourceICC,projectFolder);
```

sourceICC is the full path to an external .icc or .icm. saveICC opens Save As and makes a byte-identical copy. readICC inspects metadata; none of these calls rebuilds the profile.

Stop when an independent print meets your intended-use requirements and improvements are small relative to repeat variation and cost. Retain the previous candidate and compare using the same verification inputs.

Basis: current MATLAB functions in InkProf/src/+inkprof, checked 2026-09-29; text and print margins revised 2026-09-30. The guide describes the sequence of calls; it is not a new instrument test. Manual measurement, printing and final approval remain necessary in every actual run.

7. What does a patch role mean?

A role describes how a colour patch is used in calculation or verification.

A patch is a colour square with known RGB values and, after measurement, spectral values. Its role determines how InkProf may use that information. It is not a row, column, colour or way of scanning the instrument.

Training patches - build the profile

The measurements are used to fit the model from RGB to measured colour and create the ICC profile. A small error at these points shows how well the model fits its own inputs. It does not establish the same accuracy for new colours.

Development patches - compare alternatives

These measurements are kept out of profile fitting and used to compare, for example, smoothing and different ICC candidates. Once they guide selection, they are development data rather than a fully independent final check.

Control patches and repeats - check stability

Control patches are used for checking rather than model fitting. Identical RGB values at several positions can reveal print and measurement variation. Two scans of the same field check reading repeatability; they are not two independently printed patches.

C2 verification - test the printed result

A separate target is printed with the profile applied once and compared with the intended Lab. This assesses actual colour reproduction. If its measurements are later used to improve the model, a new separate final test is required.

Terms used in JSON and commands

RoleFile is the path to the file describing the partition. Keep it with the measurement and project manifest; changing a filename does not change how patches are used.

In C2, the role field may also indicate a colour group: gray = grayscale, colour = colour samples, challenge = deliberately difficult colours, and repeat = a repeated patch. These groups describe the verification target, not which points built the ICC profile.

Example: 80 new patches are used for training, 20 are reserved for development and 20 are repeated control patches. All 120 are measured, but only the 80 training patches are added to the profile fit.

The guide's main header, footer and text have margins of approximately 12 mm or more; the added GitHub link sits below the main footer. Print A4 at 100% without borderless enlargement. If your printer requires wider margins, choose fit to printable area.

8. Quality and ISO background

ΔE_{00} is InkProf's colour-error metric. A quality requirement also needs a reference and a test method.

Which standard applies to what?

ISO 12647-2:2013 covers offset lithographic processes. Digital contract proofs are covered by ISO 12647-7. InkProf's own RGB targets are not automatically standardised proofs, even when they use the same colour-error formula. [1, 2]

InkProf uses CIEDE2000, written ΔE_{00} . It is not the same metric as CIELAB 1976, ΔE^*_{ab} . A tolerance cannot be transferred between formulas simply by changing its label. Always state the formula, reference, patch group and whether the limit applies to the mean or maximum. [2, 3]

A limited industry example

Bundesverband Druck und Medien (bvdv), Germany's printing and media association, lists these proof tolerances in MediaStandard Print 2018, Table 30, page 50, based on ISO 12647-7:2016. [2]

Media wedge check	ΔE_{00} tolerance
Mean across all colour patches	≤ 2.5
Maximum across all colour patches	≤ 5.0
Primary-colour solids	≤ 3.0
Paper white	≤ 3.0

This is a selection of criteria, not a complete certification test. Additional criteria and prescribed measurement conditions apply. Do not transfer these limits directly to InkProf's arbitrary colour patches. A formal ISO claim requires the complete standard edition and its test procedure. [2]

Avoid blanket approvals based on a single number

"All colours below 5 are ISO-approved" and "all digital proofs must be below 1" are not valid general rules. A project-specific target such as $\Delta E_{00} \leq 1$ for important colours can be useful, but should be described as a project requirement.

This guide draws on public descriptions of standards and an industry summary. It does not replace a complete clause-by-clause review of the standards. Sources and links are on page 11.

What InkProf should report

Mean, median, P95 and maximum, the number of samples and how they were selected. P95 is the level below which approximately 95% of sample errors fall. Report grayscale and particularly important colours separately; do not let a good mean conceal local deviations.

9. Assessment and iteration

Distinguish practically acceptable final quality from small changes that help explain the model.

Approximate visual guidance - not ISO limits

ΔE_{00}	Cautious interpretation of compared colour samples
< 1.0	Very small difference; may still be visible.
1.0-2.0	Small difference, potentially visible side by side.
> 2.0-3.5	May be noticeable; acceptance depends on the image.
> 3.5-5.0	May be clear, especially for sensitive colours.
> 5.0	Investigate; the value alone does not determine acceptance.

These ranges are illustrative orders of magnitude, not universal visual thresholds. Hue, lightness, area, gloss, texture, lighting, surroundings and the observer all matter. Avoid absolute claims such as "completely invisible".

Four different error measures

Training error: how well the model fits the data used to build the profile. Development error: error on separate samples used for model selection. Print error: a physical print against its intended colour. Regression: the increase in error relative to an earlier candidate on the same samples.

Example: 0.6 becomes 0.7 ΔE_{00} . The final error is 0.7 and the increase is 0.1. If both meet the intended-use requirement, the change need not be a problem in the image. Still log it as information about the direction of iteration. The increase is not the colour distance between the candidates' colours.

Decision rules in InkProf

NormTarget=1 is a stopping target for the iteration norm, not an ISO limit. A weighted RMS norm is not the same as an arithmetic mean error. MaxPatchRegression=0.5 limits the increase in a patch error, not its absolute error to 0.5.

Assess small regressions together with final error and repeat variation. Prioritise grayscale and important colours, while tracking the whole colour range. Letting absolute quality influence the regression rule is a documented development principle; this guide introduces no new automatic selection rule.

Record paper and measurement conditions

Document the reference, illuminant/observer, M0/M1/M2, measurement geometry, backing, paper, drying time and print settings. Mark unknown information as unknown. Paper whiteness, optical brighteners and gloss affect the outcome; a small D50 error alone does not establish metamerism behaviour in other lighting.

Use repeated samples to assess variation. A small error change must not automatically be labelled noise, and estimated noise must not be subtracted from ΔE_{00} . Review measurement errors before further model optimisation.

10. Example and sources

Epson Stylus Pro 3880 / SP Glossy: profiling practically concluded, 2026-09-30.

Measured quality of the final profile

The final profile is based on 991 profiling patches. Its A2B and B2A tables are identical to Candidate-1's. The verification sheet contained 131 patches: 119 original colours and 12 repeats. L4 was subsequently replaced with an accepted spot measurement.

Check	$\Delta E00$ result
Verification: mean, 119 original patches	2.318
Verification: median	1.566
80 colours + 24 grays, no challenge: mean	1.699
Grayscale, 24 patches: mean / max	1.212 / 1.808
Verification: P95 / max, including challenge	7.927 / 11.766
Training error, 991 patches: mean	1.144

Training error is not an independent quality check. Challenge patches are deliberately difficult colours; three were selected after earlier problems. The print test supports a judgement of useful reproduction with remaining local limitations, not ISO approval.

The separately printed repeat patches differed by a mean of 0.366 and a maximum of 0.834 $\Delta E00$. The largest forward/backward scan difference was 0.150. These describe variation at different stages, not a number to subtract from profile error.

The user accepted the remaining local uncertainty and stopped further sampling. Candidate-1 is retained. An absolute physical limit of the ink has not been established. A new printer or paper requires its own measurement data.

Sources and supporting records

[1] ISO 12647-2:2013 - scope of the standard

[2] Bundesverband Druck und Medien (bvdn), MediaStandard Print 2018, Table 30, page 50

[3] CIE - Colorimetry, Part 6: CIEDE2000

The ISO overview and the table published by Bundesverband Druck und Medien (bvdn) were checked on 2026-09-30. The table concerns the stated editions, not future versions of the standards.

Project reference: docs/research/colour-difference-tolerances.md. Epson results: profile job 26df8eee-f520-4919-85e4-146170520ae0; C3 check f7a098bd-6df6-4ca4-9e63-cd038c46b719. These identifiers link the figures to saved reports and original measurements.